

01 Introduction to Machine Learning

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What is ML

Classic algorithm design: unambiguous algorithms.

Machine learning: ability to learn without being explicitly programmed.

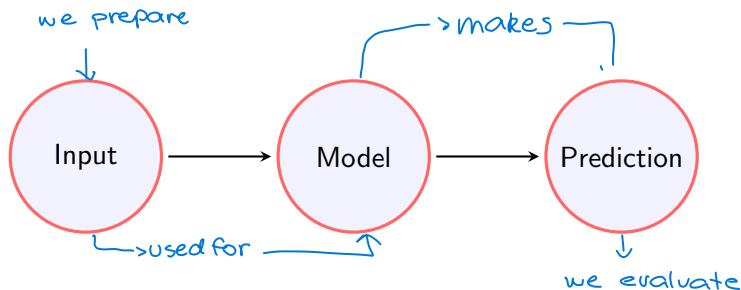
Input, model, prediction

The **model** is decoupled from the data but the algorithm is preserved.

Example: recognize lions



Input, model, predictions



Input: pre-process, analyse

Model: choose, tune hyper-parameters

Predictions: evaluate, interpret

Machine Learning Workflow

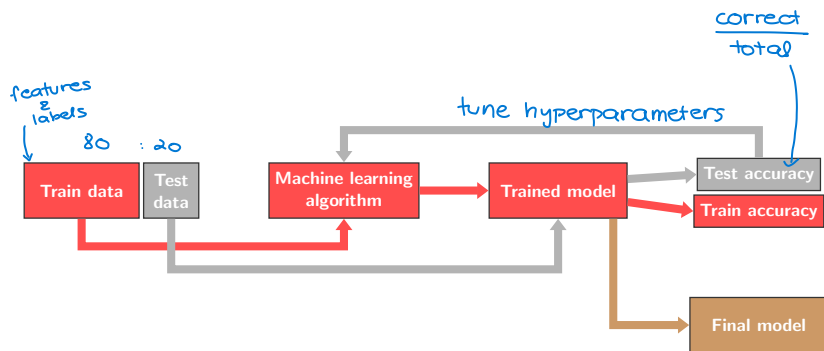


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Supervised vs. unsupervised learning

Supervised: data is labelled (we know the correct answer).
The model makes predictions for the label.

Unsupervised: data is not labelled.
The model looks for patterns in the data.

Label

- outcome
- response var.
- class
- output
- y
- dependence var.

Types of ML problems

Supervised learning: labelled data

- Classification: y is a category
- Regression: y is a number

Unsupervised learning: unlabelled data

- Clustering: identify groups in data
- Dimensionality Reduction: identify structure in data

Algorithm examples

Slido :)

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1 What is ML

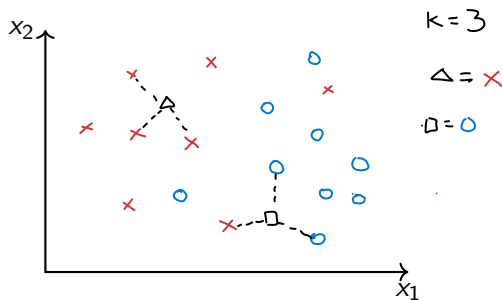
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k-Nearest neighbours

- Classification algorithm
- Finding k nearest neighbours of new data point
- Assigns majority class

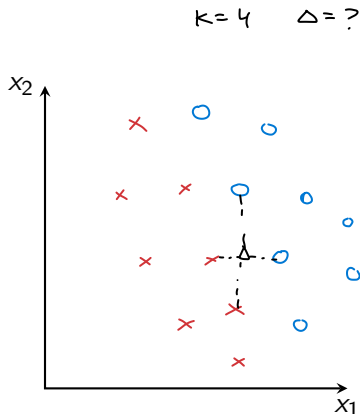


kNN hyper-parameters

- k (even/odd)
- Distance metric

Tie-breaking:

- Sum of distance
- Reduce k



Distance metric

- Euclidean •

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

- Manhattan •

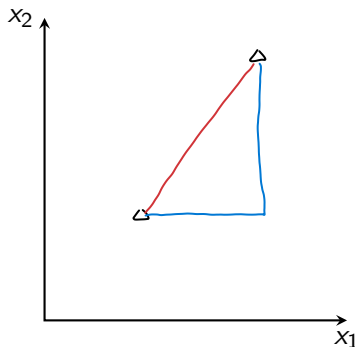
$$d(\mathbf{p}, \mathbf{q}) = \sum_{i=1}^n |p_i - q_i|$$

- Minkowski

$$d(\mathbf{p}, \mathbf{q}) = (\sum_{i=1}^n |p_i - q_i|^p)^{1/p}$$

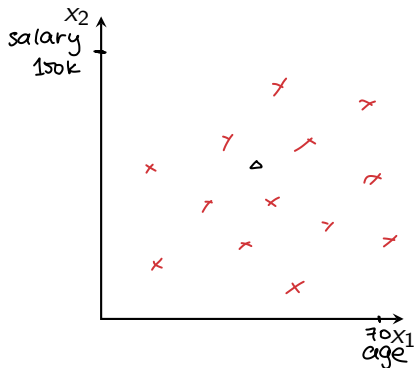
$p=1 \sim \text{Manhattan}$

$p=2 \sim \text{Euclidean}$



Scaling

In kNN, scaling matters.



salary has a much larger influence

→ scale to the same range

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Titanic example

Github link: https://github.com/304125/MAL1_S26/tree/master/01_Introduction_to_Machine_Learning